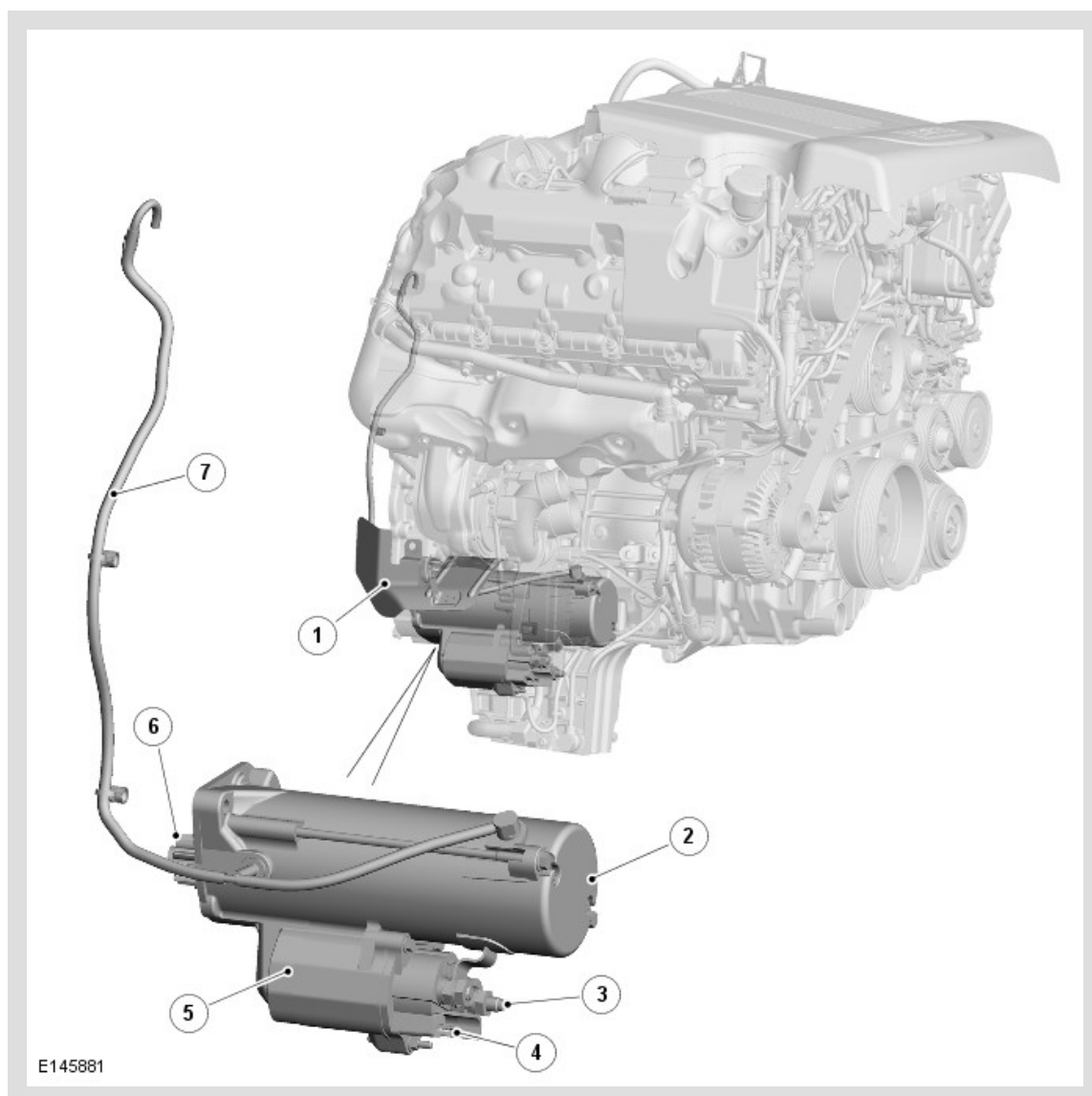


PUBLISHED: 30-SEP-2014

2015.0 RANGE ROVER (LG), 303-06

STARTING SYSTEM - TDV8 4.4L DIESEL

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Heat Shield
2	Motor
3	Motor Electrical Connection
4	Solenoid Electrical Connection
5	Solenoid
6	Pinion Gear

OVERVIEW

The starting system consists of an electric starter motor that engages with the ring gear on the crankshaft drive plate. Operation of the starter motor is controlled by the ECM (engine control module) .

DESCRIPTION

STARTER MOTOR

The starter motor is located on the rear right side of the oil pan and secured with two bolts. A heat shield is positioned above the starter motor to protect it from heat emitted from the variable turbocharger. The starter motor incorporates labyrinth-breathing tubes to help with sealing and drainage.

The starter motor is rated at 2.0 kW and comprises a series wound motor, an overrunning clutch and pinion, and a solenoid.

Power to operate the motor is supplied from the 450 A megafuse in BJB (battery junction box) 2. Power to operate the solenoid is supplied from the starter relay in the EJB (engine junction box) .

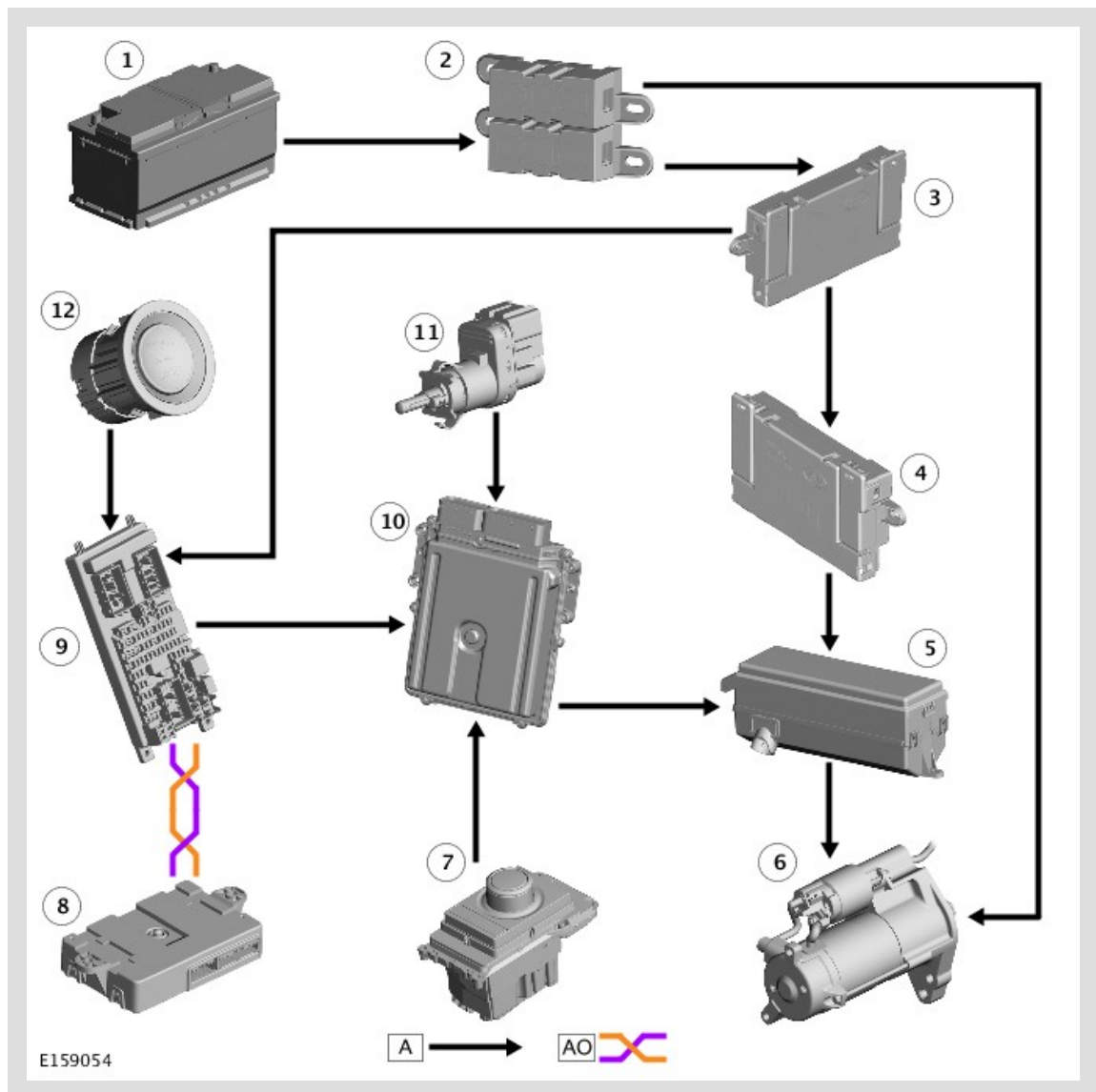
OPERATION

When the stop/start switch is pressed, if the keyless vehicle module detects a valid smart key in the vehicle the CJB (central junction box) transmits a hardwired crank request to the ECM .

For additional information, refer to: [Anti-Theft - Passive](#) (419-01B Anti-Theft - Passive, Diagnosis and Testing).

When the ECM receives a crank request from the CJB , provided the transmission control switch is in P (park) or N (neutral) and the brake pedal is pressed, it energizes the starter relay. The energized starter relay supplies battery power to the starter solenoid, which energizes and causes the pinion gear to engage with the ring gear. When the starter solenoid is energized it also closes high-current contacts for the motor, which connects battery power from BJB 2 to the motor to turn the engine.

CONTROL DIAGRAM



A = HARDWIRED; AO = MEDIUM SPEED CAN (CONTROLLER AREA NETWORK) BODY BUS.

ITEM	DESCRIPTION
1	Battery
2	BJB2 (battery junction box 2)
3	BJB (battery junction box)
4	AJB (auxiliary junction box)
5	EJB (engine junction box)
6	Starter Motor
7	TCS (transmission control switch)

8	KVM (keyless vehicle module)
9	CJB (central junction box)
10	ECM (engine control module)
11	Brake Pedal Switch
12	Stop/Start Switch